



A full-view management method based on artificial neural networks for energy and material-savings in wastewater treatment plants

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ABSTRACT

Carbon neutrality has been received extensive attention in the field of wastewater treatment. The optimal management of wastewater treatment plants (WWTPs) has great significance and urgency since the serious energy and materials waste. In this study, a full-view management method based on artificial neural networks (ANNs) for energy and material savings in WWTPs was established. More than 5 years of historical operating data from two typical plants (size 40,000 t/d and 10,000 t/d) located in Chongqing, China, were obtained, and public data in the service area of each plant were systematically collected from open channels. These abundant historical and public data were used to train two ANNs (GRA-CNN-LSTM model and PCA-BPNN model) to predict the inlets/outlets wastewater quality and quantity. The overall average prediction accuracy of inlets/outlets wastewater indicators are greater than 92.60% and 93.76%, respectively. By combining the two models, more appropriate process operation strategies can be obtained 2 weeks in advance, with more than 11.20% and 16.91% reduction of energy and material costs, respectively. This proposed method can provide full-view decision support for the optimal management of WWTPs and is also expected to support carbon emission control and carbon neutrality in the field of wastewater treatment.

1. Introduction

Greenhouse gases (GHGs) from wastewater treatment plants (WWTPs) contribute approximately 3% of total human activity emissions (IPCC et al., 2013). Among them, CH₄ emissions account for approximately 4% of the global total CH₄ emissions, which is the fifth largest emission source. The IEA "World Energy Outlook Special Report" outlines that over the next 25 years, the amount of energy used in the water sector will more than double (IEA, 2017). Therefore, in terms of global carbon emission peaks and carbon neutrality, carbon emission reduction in wastewater treatment systems is an important area of research (Song et al., 2022). Currently, most wastewater treatment processes are based on the activated sludge method, and its carbon emissions include direct and indirect emissions (mainly energy and various chemicals). In WWTPs, energy is mainly used for aeration systems, pump stations and sludge disposal processes (Li et al., 2021c), although the production and transportation of the chemicals used in the

wastewater treatment process should also be considered (Longo et al., 2016). This energy and material consumption leads to considerable indirect carbon emissions (Gma et al., 2019). However, the wastewater treatment process is a highly nonlinear system with different complex biological and physical phenomena (Jing et al., 2015; Panepinto et al., 2016), and it poses challenges for the accurate simulation and management of WWTPs. At present, most WWTPs are managed based on personal experience. To address the deteriorating water environment, improve discharge standards, and ensure effluent quality, managers of WWTPs tend to choose overaeration and chemical overdosage (Zhang et al., 2021), which inevitably consume a large amount of energy and material waste and increase carbon emissions directly or indirectly. Therefore, the optimal operation and management of wastewater treatment systems to reduce energy and material waste have become a topic of extensive research in the field of wastewater treatment.

The management of WWTPs is mainly determined by two essential elements: the inlet wastewater quality and quantity (i.e., influent load),

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and the condition of the wastewater treatment process (H. et al., 2021). The objective of the operation management at WWTPs is to reduce energy (mainly electricity) and material waste as much as possible based on the influent load and process conditions under the premise of safe discharge (Torregrossa et al., 2016). Accurate influent load information and wastewater treatment process simulation can support managers in determining a more accurate operation strategy. Therefore, the key problems of WWTP management are how to predict inlet wastewater quality and quantity and how to simulate the wastewater treatment process accurately. In recent years, rapidly developing information technologies, such as big data, artificial intelligence (AI) and the Internet of Things (IoT), have provided strong technical support and new solutions for the above challenges. The optimization of WWTP management by constructing an intelligent operation and maintenance system has become a major trend to realize energy and material savings in achieving carbon emission reductions in WWTPs. Artificial neural networks (ANNs) are a typical representative of these technologies, and they contain many specific forms, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) and back propagation neural networks (BPNNs). With a strong learning ability and outstanding predictive ability in complicated scenes, such as large data volumes and frequent data fluctuations (Cichy and Kaiser, 2019), ANNs have been widely studied in WWTPs influent load prediction and wastewater treatment process simulation.

In terms of influent load prediction, Li et al. (2019) employed autoregressive with exogenous inputs models to predict WWTP influent conditions with different hydraulic characteristics and climatic conditions. This approach effectively reduces the delay with rainfall data used as model inputs, and the calibrated model could predict wastewater flow up to 6 h ahead with good accuracy during dry weather. Dairi et al. (2019) proposed WWTP influent conditions prediction approaches based on deep learning methods and clustering algorithms to report abnormal events in advance. Results demonstrated that RNN-RBM-based OCSVM approach with an area under the curve value up to 0.98, which flagged its ability to detect the vast majority of abnormal events reported by the operator, while avoiding raising false alarms at the same time. Man et al. (2019) proposed a COD load prediction model for WWTPs based on the auto-regressive moving average (ARMA) algorithm and vector auto-regression (VAR) algorithm, and the prediction accuracy of the model reached 99%.

Regarding the wastewater treatment process simulation, Khatri et al. (2020) designed a three layered ANN model to simulate an intermittent cycle extended aeration-sequential batch reactor (ICEAS-SBR). Based on this model, the faecal coliform and total coliform removal were predicted, and the MAPE of both indicators less than 5%. Almomani (2020) proposed an ANN to simulate and predict the effluent quality of a three-stage (anaerobic/anoxic and MBBR) enhanced nutrient removal biological process (ENR-BP). The results indicated that the proposed ANN model provided effective prediction of effluent COD, TN and TP. Yaqub M et al. (2020) developed a long short-term memory (LSTM)-based ANN for predicting ammonium, TN, and TP removal efficiency in an anaerobic-anoxic-aerobic membrane bioreactor (AAO MBR) system, and the results indicated that the MSE values of ammonium, TN, and TP were 0.0047, 0.015, and 0.018, respectively. Yang et al. (2022) established a dynamic PCA-NARX hybrid model to simulate the CAST process, and the results shown that the model predicted the effluent COD and total TN with high accuracy (RMSECOD = 2.9 mg/L, RMSETN = 0.8 mg/L).

These previous studies have indicated the feasibility of predicting influent loads to WWTPs and simulating the wastewater treatment process, which provides the preconditions for data-driven optimization of wastewater treatment processes. However, in previous studies on wastewater treatment process optimization, researchers have almost always chosen only one of the predicted influent load results or wastewater treatment process simulation results as the basis for optimization, while few have combined the two to optimize the wastewater treatment

process (Li et al., 2021a; Zhao et al., 2021a; Wang et al., 2022). Yang et al. (2022) indicated that subsequent research in wastewater treatment process optimization should focus on the integration of influent loads and simulated wastewater treatment processes to accommodate dynamic load changes. The operation strategy of the wastewater treatment process is greatly affected by inlet wastewater quality and quantity. Accurate prediction of influent information will favor development of the decision support system (DSS) of WWTPs. The accurate simulation of the wastewater treatment process will enable feedback adjustment of the energy and chemical consumption in the wastewater treatment process. Combining influent loads prediction model and wastewater treatment process simulation model into a DSS for WWTPs enables more flexible operational strategies to be developed in advance, such as precise aeration, preparation of materials, rational deployment of manpower and timely maintenance of equipment. This will contribute greatly to the improvement of risk control and optimal scheduling capability of WWTPs, achieving the purpose of energy and material savings and then reducing carbon emissions.

In this study, the two parts of influent loads prediction and wastewater treatment process simulation were coupled, and a full-view management method based on ANNs for energy and material savings in WWTPs was established. Two ANN models were proposed as the core content of this full-view management method: GRA-CNN-LSTM (*gray relation analysis – convolutional neural networks – long short-term memory*) and PCA-BPNN (*principal component analysis – back propagation neural network*). The former was used to predict inlet wastewater quality and quantity, while the latter was used to simulate the wastewater treatment process. Then, the genetic algorithm was introduced to enhance the PCA-BPNN and optimize the energy and material consumption in the two WWTPs. Several similar methods are compared, indicating the effectiveness of the proposed method. This method is expected to improve the risk control ability and management level of WWTPs while saving energy and material and reducing carbon emissions.

2. Materials and methodology

As mentioned, a full-view management method based on ANNs was proposed for energy and material savings in WWTPs, and two WWTPs in Chongqing, China, were used as research objectives. They are medium and small WWTPs with scales of 100,000 m³/d and 40,000 m³/d, respectively. More than 5 years of real historical data from the two WWTPs were collected and analyzed, and a large amount of highly associated data (such as meteorological, demographic and economic data) were collected from open channels.

In this study, the inlet wastewater quality and quantity prediction model (GRA-CNN-LSTM) model was constructed, and the model was trained by a large number of real data and associated data. The model performance was compared with other commonly used model structures. Then, the wastewater treatment process simulation model (PCA-BPNN) was built and trained to express the relationship between influent, operating conditions and effluent results. Then, the GA was introduced to enhance the PCA-BPNN model. Through iterative feedback, the operating conditions under which the effluent results fall within our preset interval were obtained, which means relatively better operating conditions with lower energy and material waste. Finally, the two models GRA-CNN-LSTM and PCA-BPNN were coupled: according to the prediction results from the former, the latter can calculate a better operation strategy in advance, that is, realizing the current and future operation and maintenance. This is called a full-view management method in this study. The basic approach and model coupling process of this study are shown in Fig. 1.

2.1. Data preparation

The dataset established in this study includes historical data from two practical WWTPs and associated data in the corresponding service

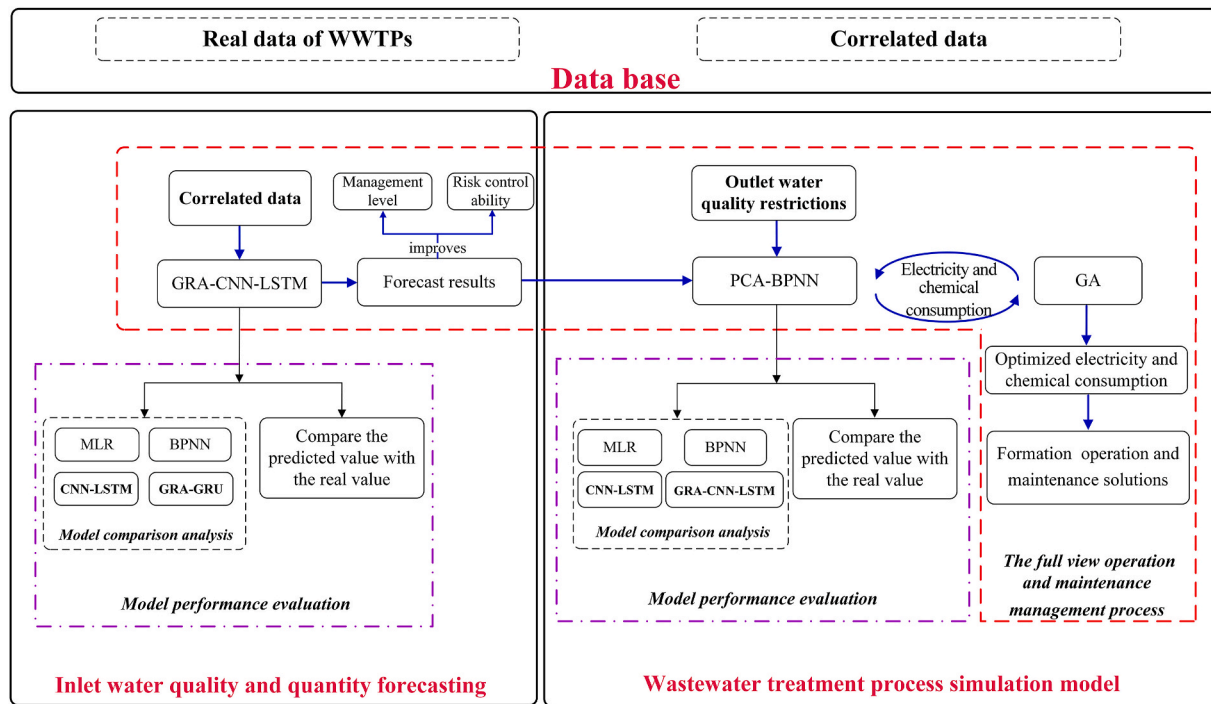


Fig. 1. The basic approach and models coupling process of this study.

area. The data collection period is from January 2016 to March 2021, a total of approximately 1900 days.

2.1.1. WWTPs historical data preparation

The two WWTPs are both located in Chongqing, China. The main processes are representative in the field of urban wastewater treatment: CAST (Cyclic Activated Sludge Technology) and OD (Oxidation Ditch) (Bo et al., 2018; Ma et al., 2018). Approximately 5 years of historical

operation and maintenance data were obtained, including inlet and outlet wastewater quality and quantity, energy consumption, sludge characteristics, and chemical dosage, for a total of approximately 30 data types.

The design capacity of the CAST process is 100,000 m³/d, with a coefficient of variation $K_Z = 1.3$. The design capacity of the OD is 40,000 m³/d, with a coefficient of variation $K_Z = 1.48$. Their effluent complies with the Class 1 A discharge standard of the “Discharge Standard of

Table 1

The actual inlet and outlet wastewater quantity and quality analysis results for CAST and OD process.

WWTPs		Indicators	Value range	Mean	Standard deviation/Removal rate
CAST process	Inlet	Water quantity (m ³)	24,294–109,546	69930.52	–
		BOD (mg/L)	24.00–386.00	89.00	37.13
		COD (mg/L)	75.00–1163.00	206.00	86.31
		SS (mg/L)	34.00–1262.0	181.00	116.46
		pH (mg/L)	7.05–9.68	7.73	0.38
		TP (mg/L)	1.36–7.70	3.49	0.98
		TN (mg/L)	14.00–43.20	29.77	5.88
	Outlet	NH ₄ ⁺ -N (mg/L)	7.54–37.10	22.48	5.74
		BOD (mg/L)	2.000–7.400	2.690	97.0%
		COD (mg/L)	5.000–24.000	11.840	94.3%
		SS (mg/L)	4.000–10.000	4.900	97.3%
		pH (mg/L)	6.710–7.710	7.150	–
		TP (mg/L)	0.050–0.250	0.140	96.0%
		TN (mg/L)	3.200–14.300	9.140	69.3%
OD process	Inlet	NH ₄ ⁺ -N (mg/L)	0.050–3.720	0.290	98.7%
		Water quantity (m ³)	17,280–50,743	36371.86	–
		BOD (mg/L)	16.00–246.00	84.00	35.32
		COD (mg/L)	71.00–791.00	284.00	116.00
		SS (mg/L)	84.00–814.00	270.00	126.66
		pH (mg/L)	7.49–8.20	7.83	0.13
		TP (mg/L)	1.14–12.40	4.93	2.21
	Outlet	TN (mg/L)	15.60–46.30	26.85	6.07
		NH ₄ ⁺ -N (mg/L)	4.31–22.70	12.18	2.69
		BOD (mg/L)	2.000–9.000	3.710	95.6%
		COD (mg/L)	6.000–24.000	11.930	95.8%
		SS (mg/L)	4.000–8.000	4.270	98.4%
		pH (mg/L)	7.290–8.110	7.640	–
		TP (mg/L)	0.030–0.290	0.100	98.0%
		TN (mg/L)	1.820–12.400	5.750	78.6%
		NH ₄ ⁺ -N (mg/L)	0.050–2.980	0.360	97.0%

Pollutants for Municipal Wastewater Treatment Plant in China" (Administration, 2002). The actual inlet and outlet wastewater quantity and quality analysis results for CAST and OD process are shown in Table 1.

2.1.2. Associated data preparation

The production of wastewater is a combination of various complex systems in society, which involves a variety of factors, including randomness, diversity, and abstraction (Harika et al.). In this study, the associated data indicate the factors that have a strong correlation with influent wastewater quality and the quantity of WWTPs. For example, the precipitation in meteorology will affect the inflow quantity, the population will affect the total amount of pollutants, and economic indicators can be used to reflect industrial wastewater discharge. According to the composition of influent wastewater, these factors can be roughly divided into three categories: economic, demographic, and meteorological. Then, the associated data were obtained from official sources, including the national statistics network, Chongqing statistics network, meteorological websites, tourism administration and government reports to ensure data stability and accuracy. Indicators closely related to wastewater quality and quantity were collected, including economic, demographic and meteorological data.

(1) Economic data

This study adopts gross regional product (E1), total import and export (E2), total retail sales of consumer goods (E3), industrial sales output (E4), total industrial output (E5), urban consumer price index (E6), urban consumer price index of food (E7), urban consumer price index of nonfood (E8), urban consumer price index of consumer products (E9), urban consumer price index of tobacco and alcoholic beverages (E10), the total industrial output value above the prescribed level (E11), the urban consumer price index of education, culture and entertainment (E12), and the sales area of commercial houses (E13) as economic data indicators. These indicators are publicly available from the government website and can directly or indirectly reflect industrial water consumption and wastewater discharge. 13 economic parameters were used because economic data were usually updated slowly and out of sync. For example, gross regional product was updated once a month, while the urban consumer price index of food was updated once a quarter. Therefore, after a preliminary exploration, these relevant parameters were chosen to compensate for each other and characterize the influence of economic factors better.

(2) Demographic data

Population will directly affect the quality and quantity of domestic wastewater. In a specific period and region, per capita water consumption and pollutant emissions are relatively stable. According to the "Standard for design of outdoor wastewater engineering" (GB 50014–2021), per capita pollutant emissions can be considered as follows: BOD₅ is 40–60 g/d, SS is 40–70 g/d, TN is 8–12 g/d, and TP is 0.9–2.5 g/d. As the "China Water Resources Bulletin in the 2019 annual" reported, the per capita comprehensive water consumption in China is 431 m³ per year. It is known that the total quantity of wastewater produced is generally 80%–90% of the water consumption. Therefore, the population data in the practical service area will favor further analysis of the inlet wastewater quality and quantity of WWTPs. This study adopts the resident population (P1) and the number of tourists (P2) as demographic data indicators.

(3) Meteorological data

Meteorological factors have a significant impact on the inlet wastewater quality and quantity and affect the operating status of WWTPs (Yan et al., 2021). According to related reports, air pressure (T1),

relative humidity (T2), rainfall (T3), temperature (T4), sunshine (T5), evaporation (T6), ground temperature (T7), and average wind speed (T8) were used as meteorological data indicators in this study.

2.2. The GRA–CNN–LSTM based inlet wastewater quality and quantity prediction model

2.2.1. The GRA algorithm

The GRA is a method of converting experimental results to a normalized value between 0 and 1. It is actually a measure of the absolute value of the difference in data between sequences and could be used to measure the approximate correlation between sequences (Hong et al., 2020).

2.2.2. The CNN algorithm

The CNN, as a recently developed algorithm, works well for data characteristics, time adaptive series data and wave data processing (Wang et al., 2019b). The general network structure of a CNN consists of an input layer, a convolution layer, a pooling layer, a fully connected layer, and an output layer (Yurtsever and Yurtsever, 2019). The network structure diagram of CNN is shown in Fig. 2.

2.2.3. The LSTM algorithm

LSTM is an improvement of the recurrent neural network (RNN) (Girshick et al., 2015), which is suitable for processing and predicting important events with long intervals and delays in a time series, and avoids the gradient disappearance and gradient explosion problems during the training of long sequences. Compared with RNN, LSTM adds a structure called a cell, within which part of the information is allowed to be kept or forgotten to determine useful information by three gates: the forget gate, the input gate, and the output gate. The network structure diagram of LSTM is shown in Fig. 3.

2.2.4. Building of the GRA–CNN–LSTM model

Too many parameter inputs will lead to an increased risk of ANN overfitting. In order to reduce the risk, GRA was used to assess the degree of influence of the parameters on the results and to adjust the parameter weights according to the degree of influence, which reduces the risk of overfitting to some extent. Hong et al. (2020) used a hybrid method of RBF neural network and GRA to effectively predict the levels of trihalomethanes in tap water. The results demonstrated that GRA is an effective method for data analysis and influence degree evaluation. CNN is a deep neural network whose weight-sharing network structure makes it more similar to a biological neural network. It can reduce the complexity of the network model, and is well suited for this study to deal with the dataset noise. Cheng et al. (2019) proposed a densely connected convolution network-based model to monitor water quality. Experimental results showed that the water quality identification rate of the model reached 99.38%. The LSTM is a time-periodic network, which is well suited to the processing of the time-series data in this study. Wang et al. (2019a) used an LSTM-based model to predict the water quality indicators of the Yellow River Basin, and the prediction accuracy was 99%. Combining CNN and LSTM neural network can process basic data with large defects, making the experiment more efficient. Wang et al. (2019b) have shown in their study of urban wastewater COD prediction that the hybrid CNN–LSTM prediction model has higher accuracy and better prediction performance compared to standalone CNN or LSTM model.

Therefore, based on the analysis of the obtained data characteristics and research objective, a GRA–CNN–LSTM model was built, and its structure is shown in Fig. 4. The steps of building the model are as follows:

In the first step, the associated data were used as input, the independent variable data in the gray relational matrix were multiplied to obtain a new independent variable matrix, and then the independent and the dependent variable matrix were normalized.

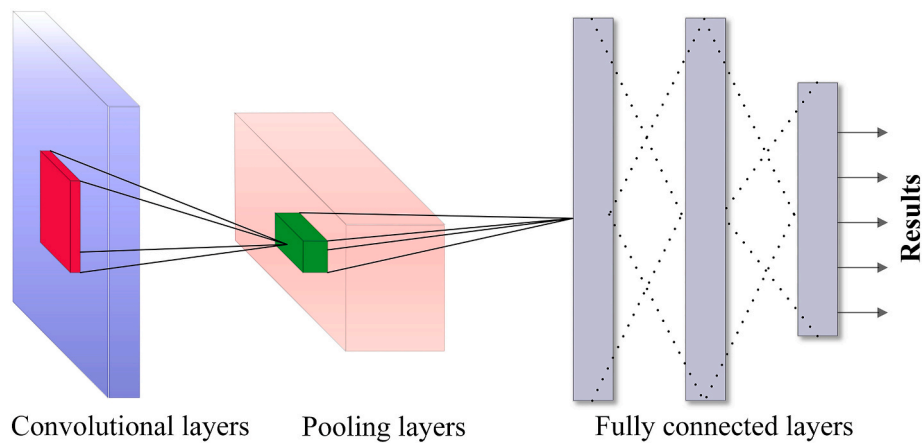


Fig. 2. Network structure diagram of CNN.

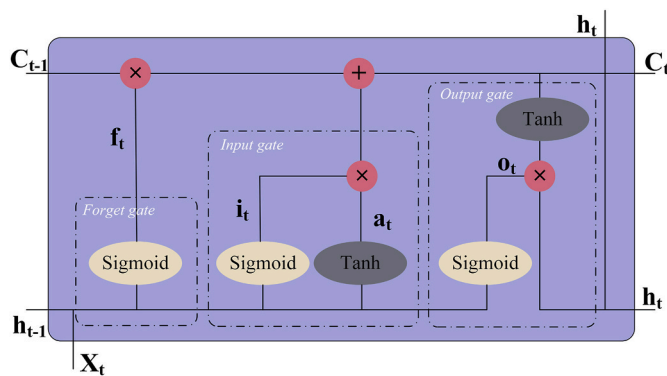


Fig. 3. Network structure diagram of the LSTM.

Second, the treated data are divided into a training set and a test set at a ratio of 3:1. The independent variables of the training and test sets were reconstructed into a three-dimensional matrix with the number of samples, time steps and several weight values. The CNN network model was established with the keras module to adjust the number and size of convolution kernels according to the actual situation, then the *Relu* activation function was adopted. Aided by the CNN, the features of the three-dimensional matrix were extracted, the pooling process was performed based on the max-pooling function that outputs a time-dependent sequence.

Finally, the sequences were used as the input of the LSTM. The input dimension of the first layer of the LSTM was 128, the input sequence was the time step and the number of samples, and the output dimension was the time step $\times$ the number of samples $\times$ the dimension of the hidden layer. The second layer of the LSTM elements had a dimension of 64, and as the last circulation layer, the test returned only the output of this step. For the training model, the number of epochs was set to 100, and the input data were trained 100 times forward and backward. The network structure table of the GRA-CNN-LSTM model is shown in Table 2, and the “None” in the table represents the number of rows, which will be adjusted according to the actual sample data.

2.3. The PCA-BPNN based wastewater treatment process simulation model

2.3.1. The PCA method

PCA is a multivariate procedure that reduces the dimensionality of a dataset that contains a large number of interrelated variables (Wang et al., 2016), and reveals the internal variation of the original variables using fewer variables (Lei et al., 2018). This method of data processing not only ensures less data loss, but also simplifies and optimizes the data in general.

2.3.2. The BPNN algorithm

The BPNN is one of the most commonly used neural networks (Bi et al., 2020). It includes input, hidden, and output layers, and it is characterized by the forward propagation of output results and

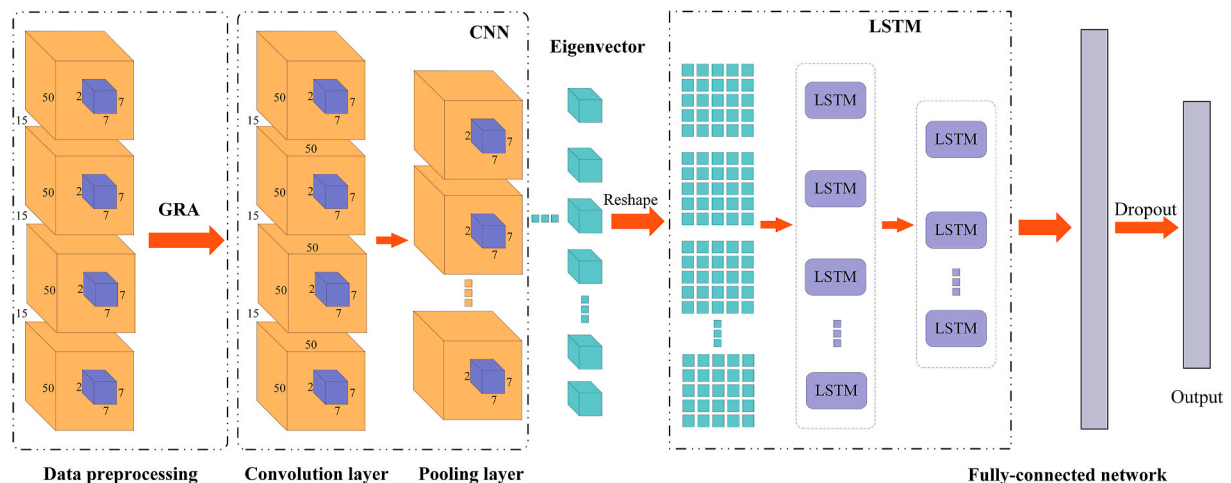


Fig. 4. The GRA-CNN-LSTM model structure.

Table 2

Network structure table of GRA-CNN-LSTM model.

No.	Layer Type	Neurons	Parameters
1	Conv1d (Conv1D)	(None,1,15)	420
2	Max_pooling_1	(None,1,15)	0
3	Lstm_1	(None,1,128)	73,728
4	Lstm_2	(None,64)	49,408
5	Dense_1 (Dense)	(None,32)	2080
6	Dense_2 (Dense)	(None,1)	33
Total params	125,669		
Trainable params	125,669		
Non-trainable params	0		

backward propagation of errors (Fei et al., 2021).

2.3.3. Building of the PCA-BPNN model

The BPNN model allows for a non-linear mapping of inputs and outputs and is suitable for multi-dimensional feature construction when performing data mining. However, the overfitting risk increases with the dimensionality of the data and neurons. Therefore, PCA was employed to pre-process the original data, simplify or remove some data while retaining the original features of the data. Deepnarain et al. (2020) used PCA and ANN tools to investigate the dominance of filamentous bacteria in seven WWTPs located in South Africa successfully. In this study, the PCA-BPNN model structure is shown in Fig. 5. The steps of building the model are as follows:

In the first step, the main features of the data were extracted by the PCA, and then the BPNN was built based on the extracted features. The fully connected neural network model consists of one input layer, two hidden layers (200 nodes per layer) and one output layer. The weights between the layers are C_{ij}^l , and the corresponding biases are δ . The activation function employs a *sigmoid* function, and the cost function between the output layer and the sample output employs a least squares function. In training, the Adam optimizer was used to solve for the minimum loss, and the optimization method was gradient descent. The number of training iterations was 10,000, and the training was stopped when the mean square error was less than 0.0001, which was considered to have reached the requirement; otherwise, the training continued until the number of training iterations reached 10,000, and the learning rate was 0.01. The model was mainly tuned by adjusting the number of hidden layers, the number of hidden layer nodes, the learning rate and the activation function. After training, the weights and biases obtained from the training were used to obtain the predicted values based on the

given sample values.

2.4. GA was introduced to enhance the PCA-BPNN model

The PCA-BPNN model established above can be used to characterize the relationship between influent, operating conditions and effluent results. The trained PCA-BPNN model can give the effluent results under a certain inflow condition and an operating condition (the data type should be consistent with the training sample). However, we hope to obtain, under a certain water inflow condition, the optimized operating condition when the effluent results meet the discharge standard. This requires the introduction of new algorithms to enhance the PCA-BPNN model. In this study, the PCA-BPNN was combined with GA to establish an energy and material consumption optimization solution for WWTPs, and this research idea has been validated and applied in other fields. Wang et al. (2021) developed a tobacco cultivation measures DSS which uses the BPNN model to simulate the tobacco cultivation process and introduces GA to enhance the model to determine cultivation measures, and the DSS has been applied in Guizhou Province, China.

2.4.1. The GA method

The GA is a search heuristic algorithm designed based on the process of genetic evolution, which is simple and general. The basic principle of the GA algorithm is that, in the evolutionary process of the population, superiority and inferiority are employed to eliminate those with low adaptive ability to the environment and keep those with relatively strong adaptive ability (Atm et al., 2019), and through this continuous selection, crossover and mutation, the optimal result is sought.

2.4.2. Introducing the GA to the PCA-BPNN model

In the specific optimization process of energy and material consumption in WWTPs, energy, iron salt, aluminum salt, desliming flocculants, high efficiency flocculant, liquid chlorine, and sodium chlorate were selected as variables, and each variable determined one or more water quality indicators of the outlet water from WWTPs. By continuously reducing the value of each variable and then using the PCA-BPNN model to predict the effluent quality, if a indicator is lower than 98% of the Class 1 A discharge standard of the Discharge Standard of Pollutants for Municipal Wastewater Treatment Plant in China, the relevant variable (such as energy, aluminum salt) will be adjusted until this indicator is between 98% and 100% of the discharge standard. Flow chart of introducing GA into PCA-BPNN model is shown in Fig. 6.

The main parameters of the GA are the number of populations and

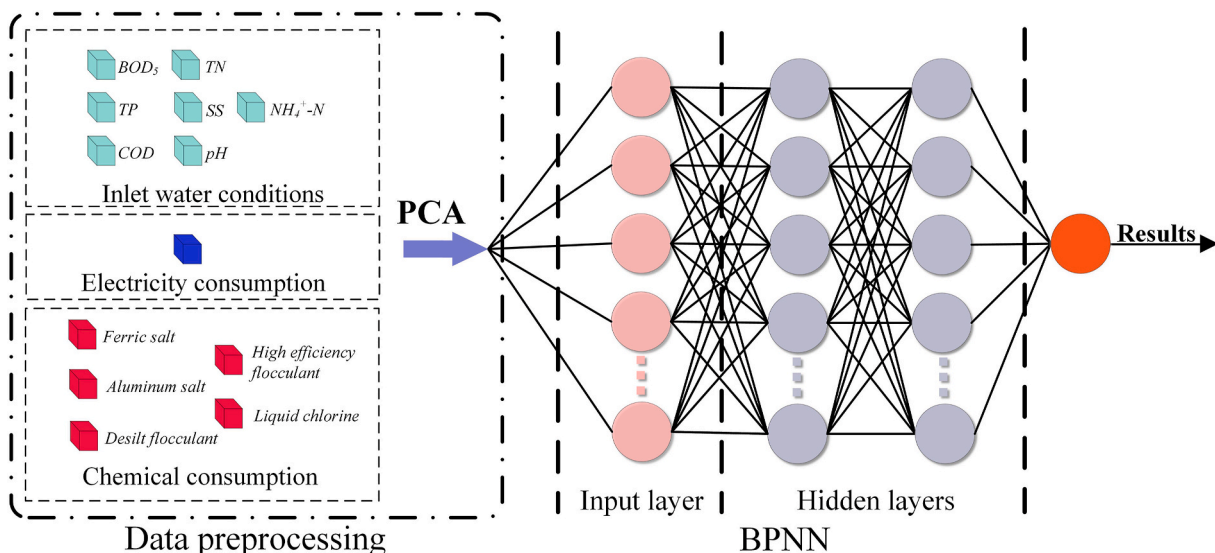


Fig. 5. The PCA-BPNN model structure.

the function. In this study, the maximum and minimum values of the variables were encoded according to the binary numbers, and then the initialized populations were randomly generated, where the initialization produces 100 populations, and each population generated binary code and created the adaptation function, according to the saved weights and biases. The binary code was decoded and input into the adaptation function, and the fitness of the populations was judged: those satisfying the conditions were retained, and those that did not satisfy the conditions were removed. The values of indicators such as energy were cross-acted, and the resulting new populations were coded. Then, the previous step was repeated until an optimal solution was generated that satisfied the conditions.

2.5. Model evaluation methodologies

In this study, the performance of the GRA-CNN-LSTM model and the PCA-BPNN model was evaluated using the mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE). The MAE, RMSE, and MAPE scores are closer to 0, indicating a better model fitting effect.

$$MAE(\varnothing, \varnothing) = \frac{1}{\vartheta} \sum_{i=1}^{\vartheta} |\varnothing_i - \varnothing_n| \quad (2-24)$$

$$RMSE = \sqrt{\frac{1}{\vartheta} \sum_{i=1}^{\vartheta} (\varnothing_i - \varnothing_n)^2} \quad (2-25)$$

$$MAPE = \frac{1}{\vartheta} \sum_{i=1}^{\vartheta} \frac{|\varnothing_i - \varnothing_n|}{\varnothing_i} \quad (2-26)$$

where $\varnothing_i$ represents the actual value, $\varnothing_n$ represents the predicted value, and ϑ represents the number on the sample dataset.

2.6. Comparison model

Different model structures and types have their corresponding advantages and scenarios. Based on the analysis of the obtained data and the research objective, this study built two neural network models. To compare their performance, several representative methods were selected from related reports (Liu et al., 2018; Zhang et al., 2018; Ayzel and Heistermann, 2021; Barzegar et al., 2021) in the fields of wastewater treatment and watershed treatment in recent years.

MLR: MLR is a statistical analysis method that uses mathematical

knowledge to determine quantitative relationships between variables, and it is applied to two or more variables, and MLR has frequently been used to compare with ANN models in many previous studies (Çerçi and Hürdoğan, 2020; Bera et al., 2021; Rao et al., 2021; Shams et al., 2021).

BPNN: A BPNN is a multilayer feed-forward neural network with a simple and clear model architecture, simple computation, the ability to learn and store multiple mapping relationships (Fei et al., 2021), and is widely used in the field of prediction.

CNN-LSTM: Compared with the GRA-CNN-LSTM model, the CNN-LSTM model omits the data cleaning operation, and the original data directly enter the CNN-LSTM model for training. Wang et al. (2019b) constructed a CNN-LSTM model to predict municipal wastewater COD concentrations, and without GRA analysis of the data, their analysis suggested that the predictive performance of the method was adequate for industrial requirements and could be applied to municipal WWTPs.

GRA-GRU: The gated recurrent unit (GRU) is able to autonomously obtain different variables at different times for each recurrent unit, which can solve the problem of gradient disappearance of the traditional recurrent neural network model and is a recurrent neural network model such as LSTM (Ayzel and Heistermann, 2021). The advantage of GRU is fewer parameters and shorter running time. Li et al. (2021b) proposed a GRU-based method for predicting dissolved oxygen in fisheries and achieved significant results, which showed that evaluation indicators MAE, MSE, MAPE, and R^2 of GRU are 0.450 mg/L, 0.411, 0.054, and 0.994. In this application scenario, the authors also compared the performance of GRU with LSTM, and the results shown that the performance of GRU was similar to the LSTM, but the time cost and parameters number used by GRU was much less than LSTM.

3. Results and discussion

3.1. Inlet wastewater quality and quantity prediction

3.1.1. Performance analysis of the GRA-CNN-LSTM model

This study built a GRA-CNN-LSTM model to predict the inlet wastewater quality and quantity of the two WWTPs. Approximately 5 years of historical data from the WWTPs and the associated data (economic data, demographic data, and meteorological data) in the corresponding service area were used to train and test the model. The model's prediction performance is shown in Table 3 and Table 4.

As shown above, the MAE, RMSE, and MAPE of the GRA-CNN-LSTM model for the 8 indicators of the two WWTPs are all lower than 1. Considering the model accuracy, this shows that the model has a certain feasibility to be applied to the prediction of inlet wastewater quality and quantity. To further demonstrate the superiority of the GRA-CNN-LSTM model, four other widely used models (MLR, BPNN, CNN-LSTM and GRA-GRU) were established and were also trained and tested with the same training algorithm to optimize for estimating the targets. The prediction performance of these models was compared with the GRA-CNN-LSTM model, and the comparison results are shown in Fig. 7 and Fig. 8.

According to the comparison results, it is clear that the MAE, RMSE and MAPE values of the GRA-CNN-LSTM model are lower than those of the other comparison models, which indicates to some extent the

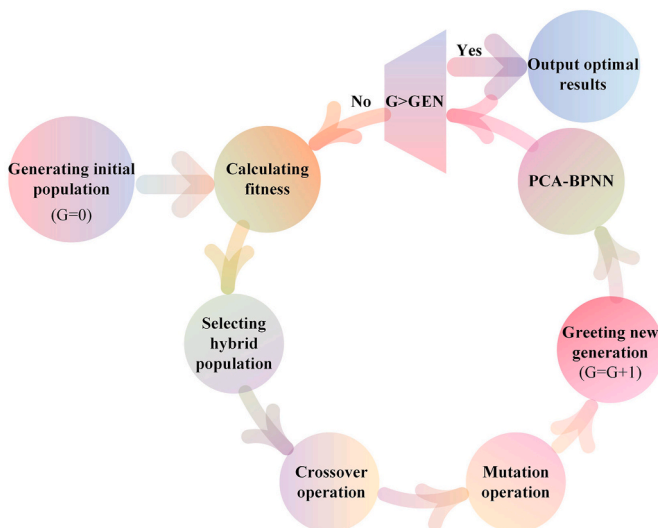


Fig. 6. Flow chart of introducing GA into PCA-BPNN model.

Table 3

The GRA-CNN-LSTM model evaluation indicators (CAST process).

Objectives	MAE	RMSE	MAPE
Inlet wastewater quantity (m^3)	0.590	0.850	0.260
Inlet wastewater BOD_5 (mg/L)	0.136	0.470	0.650
Inlet wastewater COD (mg/L)	0.157	0.340	0.540
Inlet wastewater SS (mg/L)	0.138	0.526	0.870
Inlet wastewater pH	0.218	0.360	0.590
Inlet wastewater TP (mg/L)	0.174	0.690	0.340
Inlet wastewater TN (mg/L)	0.320	0.270	0.200
Inlet wastewater $NH_4^+ - N$ (mg/L)	0.280	0.360	0.300

Table 4

The GRA-CNN-LSTM model evaluation indicators (OD process).

Objectives	MAE	RMSE	MAPE
Inlet wastewater quantity (m^3)	0.23	0.33	0.12
Inlet wastewater BOD_5 (mg/L)	0.48	0.69	0.42
Inlet wastewater COD (mg/L)	0.24	0.18	0.88
Inlet wastewater SS (mg/L)	0.69	0.39	0.68
Inlet wastewater pH	0.50	0.48	0.33
Inlet wastewater TP (mg/L)	0.34	0.29	0.67
Inlet wastewater TN (mg/L)	0.11	0.58	0.31
Inlet wastewater $NH_4^+ - N$ (mg/L)	0.146	0.26	0.27

superiority of the GRA-CNN-LSTM model in the inlet wastewater quality and quantity prediction of WWTPs. Further comparisons found that the MAE, RMSE and MAPE values of the single model were significantly higher than those of the hybrid model, and these evaluation indicators of the CNN-LSTM model were higher than those of the GRA-GRU and GRA-CNN-LSTM models. The comparison results showed that the high accuracy of the GRA-CNN-LSTM can be attributed to the combination of the advantages of the hybrid model and GRA. Meanwhile, the higher prediction accuracy of the GRA-CNN-LSTM model showed that the selected associated data can effectively reflect the daily inlet wastewater quality and quantity of WWTPs.

3.1.2. Prediction results analysis of the GRA-CNN-LSTM model

In this study, the inlet wastewater quality and quantity of the two WWTPs were predicted by the GRA-CNN-LSTM model. The predicted values were compared with the actual values to analyze the prediction performance of the model. As representative results, the predicted and actual inlet wastewater quality and quantity of February (the month with the lowest average temperature) and August (the month with the highest average temperature) in 2019 are shown in Fig. S1, and Fig. S2.

According to the comparison result figures, it was found that the predicted values of the GRA-CNN-LSTM model for the two WWTPs have

approximately the same trend as the actual values, which further revealed that the associated data selected in this study can effectively reflect the daily inlet wastewater quality and water quantity of WWTPs.

For further analysis, the predicted value accuracy of the GRA-CNN-LSTM model for each indicator of the two WWTPs is shown in Table 5 and Table 6. According to the results, the GRA-CNN-LSTM model exhibited high accuracy for the two WWTPs. Among them, the maximum accuracy reached approximately 99%, and the average accuracy value was in the range of 90.57%–98.00%. The overall average prediction accuracy of the two WWTPs is 92.60%. However, in the prediction of BOD_5 , COD and SS indicators, although the maximum and average accuracies are relatively high, the minimum accuracies are not satisfactory (the average minimum accuracies were all less than 80%). The reason may be the interference of some data noise, including data

Table 5

The GRA-CNN-LSTM model accuracy statistics of the real value and predicted value (CAST process).

Objectives	Maximum accuracy	Minimum accuracy	Average accuracy
Inlet wastewater quantity (m^3)	99.98%	88.49%	92.12%
Inlet wastewater BOD_5 (mg/L)	99.92%	80.27%	92.19%
Inlet wastewater COD (mg/L)	99.17%	71.29%	90.57%
Inlet wastewater SS (mg/L)	99.99%	70.56%	91.28%
Inlet wastewater pH	99.96%	95.17%	98.00%
Inlet wastewater TP (mg/L)	99.78%	81.71%	91.01%
Inlet wastewater TN (mg/L)	99.13%	81.69%	91.33%
Inlet wastewater $NH_4^+ - N$ (mg/L)	98.96%	85.00%	92.01%

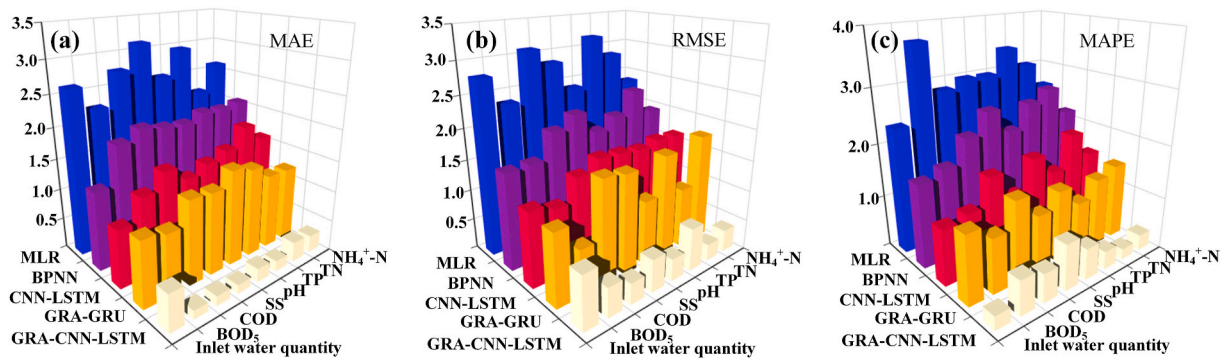
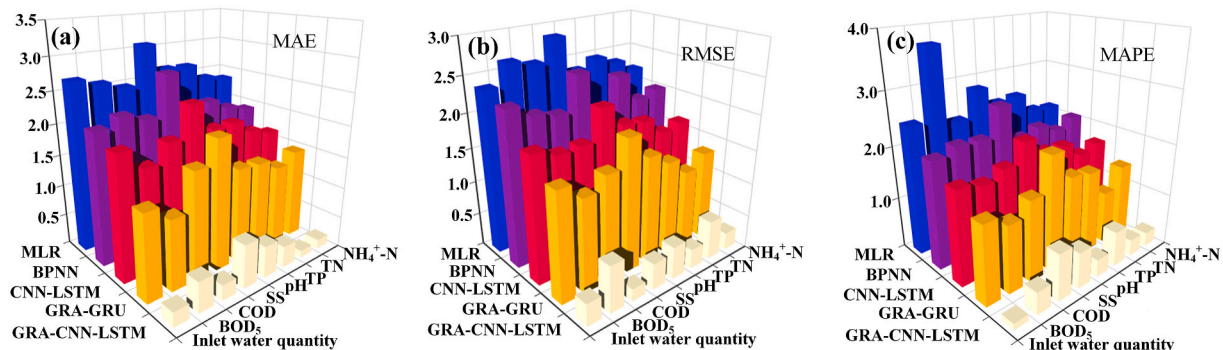
**Fig. 7.** Performance comparison of the GRA-CNN-LSTM model with different models (CAST process).**Fig. 8.** Performance comparison of the GRA-CNN-LSTM model with different models (OD process).

Table 6

The GRA-CNN-LSTM model accuracy statistics of the real value and predicted value (OD process).

Objectives	Maximum accuracy	Minimum accuracy	Average accuracy
Inlet wastewater quantity (m ³)	99.97%	89.65%	96.51%
Inlet wastewater <i>BOD</i> ₅ (mg/L)	99.92%	70.23%	91.86%
Inlet wastewater <i>COD</i> (mg/L)	99.76%	75.10%	91.24%
Inlet wastewater <i>SS</i> (mg/L)	99.84%	78.53%	92.05%
Inlet wastewater <i>pH</i>	99.96%	89.33%	97.60%
Inlet wastewater <i>TP</i> (mg/L)	99.60%	80.08%	91.48%
Inlet wastewater <i>TN</i> (mg/L)	99.71%	80.10%	90.85%
Inlet wastewater <i>NH</i> ₄ ⁺ - <i>N</i> (mg/L)	99.90%	85.00%	91.52%

bias in correlated data and historical data of WWTPs. Statistical analysis of the fluctuations in prediction accuracy for each indicator shows that *BOD*₅, *COD* and *SS* have the largest fluctuations in prediction and had similar distribution patterns ($87.59\% \pm 12.34\%$, $86.33\% \pm 13.14\%$ and $87.23\% \pm 12.69\%$, respectively). This may be due to the inclusion of some particulate *BOD*₅ and *COD* in the *SS*, resulting in an implied correlation between the three indicators, which also makes prediction more difficult. Similarly, *TP* and *TN* have similar predicted fluctuation distributions ($90.29\% \pm 9.40\%$ and $90.16\% \pm 9.26\%$, respectively), which may be due to the fact that in wastewater plants they were monitored with the same multifunctional online monitoring instrument and the machine measurement errors were relatively consistent. The *pH* has the highest accuracy and the smallest fluctuations in prediction ($96.11\% \pm 3.86\%$) due to the relatively small variation in municipal domestic wastewater. Wastewater quantity and *NH*₄⁺-*N* are predicted with accuracy and volatility ($94.52\% \pm 5.45\%$ and $92.22\% \pm 7.22\%$, respectively) second only to *pH*. In summary, the prediction method of WWTPs inlet wastewater quality and quantity proposed in this study has high prediction accuracy, and it is valuable for engineering guidance and reference.

3.2. The wastewater treatment process simulation

3.2.1. Performance analysis of the PCA-BPNN model

In this study, a PCA-BPNN model was built to simulate the wastewater treatment process of the two WWTPs. Approximately 5 years of historical data, including energy consumption, material consumption, and inlet wastewater quantity and quality (*BOD*₅, *COD*, *SS*, *pH*, *TP*, *TN* and *NH*₄⁺ - *N*), were used to train and test the model. The prediction performance is shown in Table 7 and Table 8.

As shown above, the MAE, RMSE, and MAPE of the PCA-BPNN model for these 7 indicators of the two WWTPs are all lower than 1. Considering the model accuracy, it shows that the model has a certain feasibility to be applied in the simulation of the wastewater treatment process. Meanwhile, to further analyze the performance of the PCA-

Table 7

The PCA-BPNN model evaluation indicators (CAST process).

Objectives	MAE	RMSE	MAPE
Outlet water <i>BOD</i> ₅ (mg/L)	0.340	0.150	0.250
Outlet water <i>COD</i> (mg/L)	0.130	0.260	0.240
Outlet water <i>SS</i> (mg/L)	0.100	0.140	0.260
Outlet water <i>pH</i>	0.097	0.098	0.100
Outlet water <i>TP</i> (mg/L)	0.120	0.098	1.300
Outlet water <i>TN</i> (mg/L)	0.170	0.120	0.140
Outlet water <i>NH</i> ₄ ⁺ - <i>N</i> (mg/L)	0.160	0.130	0.150

Table 8

The PCA-BPNN model evaluation indicators (OD process).

Objectives	MAE	RMSE	MAPE
Outlet water <i>BOD</i> ₅ (mg/L)	0.49	0.17	0.36
Outlet water <i>COD</i> (mg/L)	0.19	0.15	0.20
Outlet water <i>SS</i> (mg/L)	0.13	0.17	0.37
Outlet water <i>pH</i>	0.52	0.46	0.54
Outlet water <i>TP</i> (mg/L)	0.19	0.21	0.27
Outlet water <i>TN</i> (mg/L)	0.29	0.39	0.44
Outlet water <i>NH</i> ₄ ⁺ - <i>N</i> (mg/L)	0.24	0.18	0.30

BPNN model, 4 representative models (MLR, BPNN, CNN-LSTM and GRA-CNN-LSTM) were established were also trained and tested with the same training algorithm to optimize for estimating the targets. The prediction performance of these models was compared with the PCA-BPNN model, as shown in Fig. 9 and Fig. 10.

According to the comparison results, it is clear that the MAE, RMSE and MAPE values of the GRA-CNN-LSTM model are lower than those of the other models, which indicates its superiority in this application scenario. Meanwhile, it was found that the MAE, RMSE and MAPE values of the MLR and BPNN models were much higher than those of the other hybrid models. This result again showed that hybrid models were more advantageous than individual models in this scenario.

3.2.2. Analysis of simulation results of the PCA-BPNN model

Similar to the prediction results analysis of the GRA-CNN-LSTM model, the simulation values of the PCA-BPNN model were also compared with the actual values to analyze its performance. As representative results, the simulation and actual effluent quality of February (the month with the lowest average temperature) and August (the month with the highest average temperature) of 2019 are shown in Fig. S3, and Fig. S4. According to those comparison figures, the simulation value has a similar trend to the real value and has a better goodness-of-fit.

To further analyze the performance of the PCA-BPNN model, the simulation accuracies of the two WWTPs are shown in Table 9, and Table 10. The average accuracy is above 90.00%, among which the highest average accuracy is 98.30% for simulating the effluent *SS* value. Statistical analysis of the fluctuations in the prediction accuracy of each indicator ($91.12\% \pm 8.66\%$, $90.06\% \pm 9.86\%$, $92.45\% \pm 7.40\%$, $94.49\% \pm 5.37\%$, $90.76\% \pm 9.14\%$, $90.43\% \pm 9.54\%$ and $91.68\% \pm 7.50\%$ for *BOD*₅, *COD*, *SS*, *pH*, *TP*, *TN* and *NH*₄⁺-*N* respectively) shows that they all have small fluctuations in forecast (less than 10%) and a high forecast accuracy. Similar to the predicted results for inlet wastewater quality and quantity, the *pH* has the lowest prediction fluctuations and the highest prediction accuracy. The overall average prediction accuracy of the two WWTPs is 93.76%.

The prediction of outlet wastewater quality is significantly better than the prediction of inlet wastewater quality. The possible reason is the historical WWTPs data is more accurate and stable than the associated data. Overall, the accuracy and stability were maintained at a high level, demonstrating the superiority of the PCA-BPNN model. The simulation results have effective reference and decision-making value for the management of WWTPs.

3.3. Analysis of energy and material optimization results

By analyzing the dataset, it was found that the influents *BOD*₅, *COD* and *SS* fluctuated relatively widely. Therefore, in this study, these three indicators were arranged in descending order, and 10 sets of representative historical inlet wastewater conditions from the two WWTPs were selected for simulation and analysis. With these inlet wastewater conditions, the energy and material consumption were optimized by PCA-BPNN and GA under the premise of ensuring the effluent quality to meet the discharge standard. The optimization results of energy and material consumption for the two WWTPs are shown in Fig. 11 and

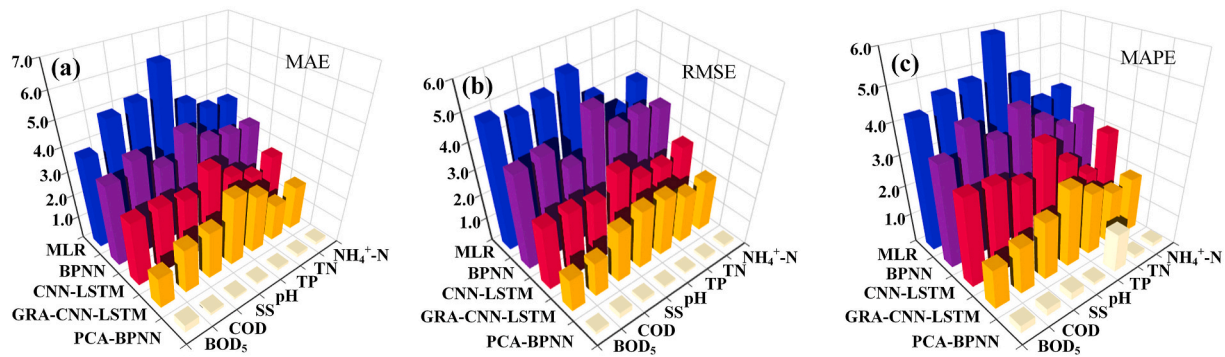


Fig. 9. Performance comparison of the PCA-BPNN model with different models (CAST process).

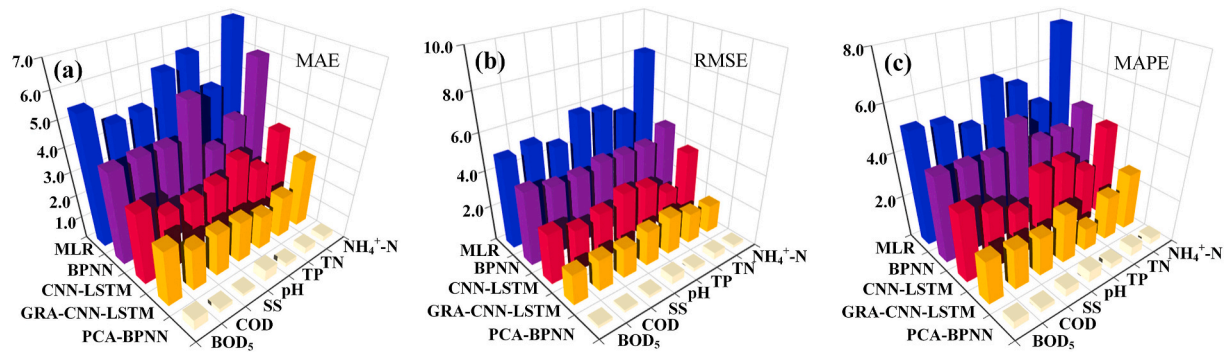


Fig. 10. Performance comparison of the PCA-BPNN model with different models (OD process).

Table 9
The PCA-BPNN model accuracy statistics (CAST process).

Objectives	Maximum accuracy	Minimum accuracy	Average accuracy
Outlet water BOD_5 (mg/L)	99.91%	84.14%	97.27%
Outlet water COD (mg/L)	99.94%	80.13%	90.99%
Outlet water SS (mg/L)	99.70%	80.17%	93.43%
Outlet water pH	99.72%	86.64%	94.62%
Outlet water TP (mg/L)	99.96%	82.48%	93.64%
Outlet water TN (mg/L)	99.94%	80.15%	92.17%
Outlet water $NH_4^+ - N$ (mg/L)	99.05%	83.07%	95.88%

Table 10
The PCA-BPNN model accuracy statistics (OD process).

Objectives	Maximum accuracy	Minimum accuracy	Average accuracy
Outlet water BOD_5 (mg/L)	99.65%	80.77%	91.70%
Outlet water COD (mg/L)	99.90%	80.26%	92.26%
Outlet water SS (mg/L)	99.99%	89.94%	98.30%
Outlet water pH	99.99%	91.60%	91.53%
Outlet water TP (mg/L)	99.83%	80.76%	92.67%
Outlet water TN (mg/L)	100.00%	81.63%	94.44%
Outlet water $NH_4^+ - N$ (mg/L)	99.32%	85.29%	93.69%

Fig. 12.

According to optimization results, with the simulation and optimization method proposed in this study, the WWTP with the CAST process could save 1%–48% of energy, 3%–26% of Ferric salt, 2%–27% of

aluminum salt, 7%–30% of desilt flocculant, 4%–35% of high efficiency flocculant, and 5%–60% of liquid chlorine. The average energy and materials saving rate were 14.99% and 18.19%, respectively. The WWTP with the OD process could save 1%–18% of energy, 2%–32% of phosphorus remover, 15%–44% of hydrochloric acid, 5%–25% of sodium chlorate, 10%–32% of flocculants, and 7%–30% of glucose. The average energy and materials saving rate of the OD process were 11.20% and 16.91%, respectively. The effluent quality given by this method is all below or close to the discharge standard. The above results verified that the energy and material consumption of the wastewater treatment process can be optimized by the GA under artificially set conditions. This feature can meet some further demand in practical engineering applications. To ensure effluent safety, most WWTPs set a series of internal discharge standards, such as 70% of the standards required by the government. The method proposed in this study allows users to set target values flexibly according to their needs.

3.4. Formation of full-view management solutions

By coupling the GRA-CNN-LSTM and PCA-BPNN models, the inlet wastewater quality and quantity can be predicted in advance, and the wastewater treatment process can be simulated. With the aid of the GA, energy and material optimization results of the process can be obtained. Then a comprehensive perspective and management method was established to realize the current and future operation and maintenance. What needs to be discussed next is how these results can be applied to a certain process management. To obtain specific operation management strategies, it is necessary to allocate information on energy-consuming equipment and the material-consuming process section. Based on the results of the GA optimization of energy consumption, this study formed a set of equipment operation plans under certain influent conditions following the following principles: A. reducing the number of pump starts and stops; B. equalizing the accumulated time of each pump; and C. coordinating the main power-consuming equipment.

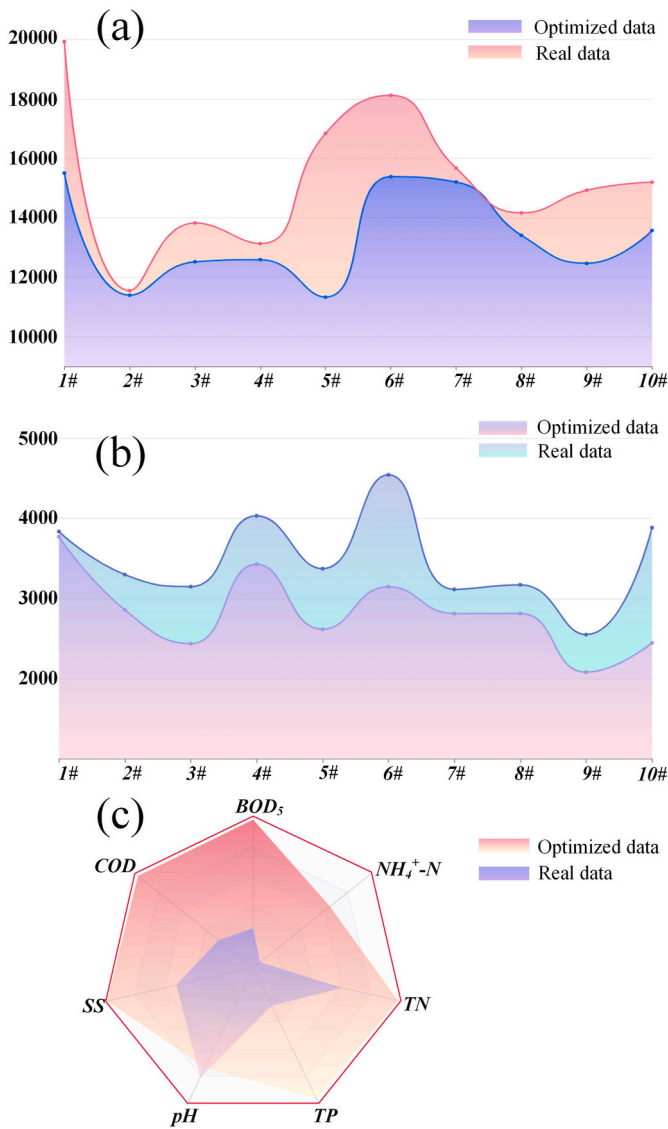


Fig. 11. (a) Electricity consumption optimization results, (b) chemical consumption optimization results, (c) outlet water quality optimization results. (CAST process).

According to the investigation of the two WWTPs, the main electrical consumption devices are shown in Table 11 and Table 12. According to the above different influent conditions, the corresponding operation strategies were proposed. The results are shown in Table S1 and Table S2, respectively.

4. Conclusions

A full-view management method for energy and material savings in WWTPs was built in this study, and its core is two neural networks: GRA-CNN-LSTM and PCA-BPNN. The effect of this method was compared with the data from the two WWTPs, and main conclusions are as follows:

(1) The proposed GRA-CNN-LSTM model performed well in predicting the inlet wastewater quality and quantity of WWTPs, and its average prediction accuracy ranged from 90.57% to 98.00% for each indicator, and the overall average prediction accuracy of the two WWTPs is 92.60%. Compared with the other four representative models, the superiority of the GRA-CNN-LSTM model in this application scenario was verified.

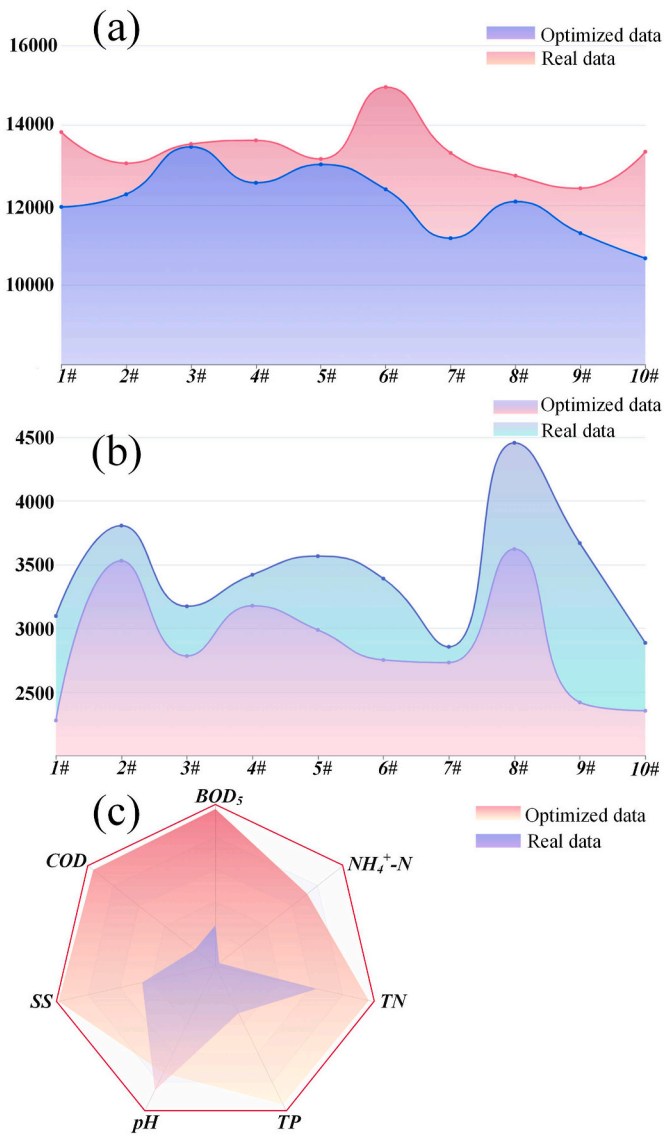


Fig. 12. (a) Electricity consumption optimization results, (b) chemical consumption optimization results, (c) outlet water quality optimization results. (OD process).

Table 11
Major power consuming equipment (CAST process).

May-November	Number	Power (kW)	Opening Value
Blowers	3	160	0.9
Lifting pumps	2	85	0.65
Deslimer	2	15	1
Axial pumps	3	30	0.95
December-April	Number	Power (kW)	Opening Value
Blowers	4	160	0.9
Lifting pumps	2	85	0.65
Deslimer	2	15	1
Axial pumps	2	30	0.95

(2) The PCA-BPNN model was successfully built to simulate the wastewater treatment process and predict the outlet wastewater quality of WWTPs. Its average prediction accuracy ranges from 90.99% to 98.30% for each indicator, and the overall average prediction accuracy is 93.76%. Compared with the other four representative models, the superiority of the PCA-BPNN model in this application scenario was verified.

Table 12

Major power consuming equipment (OD process).

	Number	Power (kW)	Opening Value
Blowers	3	160	0.95
Lifting pumps	2	85	0.8
Deslimmer	2	18	1
Axial pumps	3	30	0.95

- (3) A GA was introduced to enhance the PCA-BPNN model, and the operation strategies were generated with lower energy and material consumption: The average energy and materials saving rate of the CAST process were 14.99% and 18.19%; The average energy and materials saving rate of the OD process were 11.20% and 16.91%, respectively.
- (4) By coupling the two neural network models, a full-view management method was established to realize the current and future operation and maintenance. A more accurate operation management strategy of the sewage treatment plant can be formulated 2 weeks in advance. For the managers, manpower scheduling, equipment maintenance and material preparation can be carried out in advance, which can effectively improve the risk response ability and management level of the WWTPs. These results indicate that the associated data selected in this study are reliable and the model is valid and has a degree of engineering guidance and reference significance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2022.113054>.

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